

Machine learning methods to assist energy system optimization

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HIGHLIGHTS

- A surrogate is developed using supervised and transfer learning.
- Surrogate model is coupled with the optimization algorithm to design energy systems.
- Model adaptation allows the tool to be used even with notable changes in demand and generation.
- The computational time can be reduced up to 84%.

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ABSTRACT

This study evaluates the potential of supervised and transfer learning techniques to assist energy system optimization. A surrogate model is developed with the support of a supervised learning technique (by using artificial neural network) in order to bypass computationally intensive Actual Engineering Model (AEM). Eight different neural network architectures are considered in the process of developing the surrogate model. Subsequently, a hybrid optimization algorithm (HOA) is developed combining Surrogate and AEM in order to speed up the optimization process while maintaining the accuracy. Pareto optimization is conducted considering Net Present Value and Grid Integration level as the objective functions. Transfer learning is used to adapt the surrogate model (trained using supervised learning technique) for different scenarios where solar energy potential, wind speed and energy demand are notably different. Results reveal that the surrogate model can reach to Pareto solutions with a higher accuracy when grid interactions are above 10% (with reasonable differences in the decision space variables). HOA can reach to Pareto solutions (similar to the solutions obtained using AEM) around 17 times faster than AEM. The Surrogate Models developed using Transfer Learning (SMTL) shows a similar capability. SMTL combined with the optimization algorithm can predict Pareto fronts efficiently even when there are significant changes in the initial conditions. Therefore, STML can be used along with the HOA, which reduces the computational time required for energy system optimization by 84%. Such a significant reduction in computational time enables the approach to be used for energy system optimization at regional or national scale.

1. Introduction

Distributed energy systems can play a vital role when integrating SPV and wind energy technologies. A number of different concepts such as virtual plants, smart micro grids, energy hubs, integrated energy systems etc. are emerging within the umbrella of distributed energy systems due to its capabilities to integrate non-dispatchable energy technologies with a minimum impact to the grid [1,2]. Distributed energy systems integrate system component which possess different

characteristics, enabling to cater the demand reliably during the periods with lower wind and solar energy potential. Devices with different characteristics may result in a complex energy flow within the energy system [3]. As a result, dispatch optimization needs to be considered during the design optimization of the energy system. Considering both system design and operation strategy during the optimization process makes the energy system optimization a challenging task [4]. Uncertainties in energy demand, renewable energy potential, market prices of the system components etc. further add to the difficulty [5,6].

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Nomenclature

$\eta^{\text{W-losses}}$	losses that take place in wind turbines	MILP	mixed integer linear programming
θ	cell temperature	OMF	fixed operating and maintenance
A^{SPV}	represents the area of a single	OMV	variable operating and maintenance cost
AEM	actual Engineering Model	p	real interest rate
AM	air-mass	$P_t^{\text{Bat. max}}$	maximum power that can be taken from the battery bank
AR	architecture of the neural network	$P_t^{\text{FG max}}$	maximum power that can be taken from the grid
CFD	computational fluid dynamics	P^{ngen}	power generation capacity (maximum)
CRF	capital recovery factor	P_R	rated power of the wind turbine
ELD_t	electricity load demand	PTG	power units sell back to the grid
G_t^β	global-solar irradiation	SGD	stochastic gradient decent
GCF	price of grid electricity when purchasing electricity	SMTL	surrogate model trained using transfer learning
GCT	price of grid electricity when selling back	SPV	solar PV scenario
HOA	hybrid optimization algorithm	SVM	support vector machines
ICC	initial capital cost	v_{CI}	cut-in wind speed
ICG	internal combustion generator	v_{CO}	cut-off wind speed
MAE	mean absolute error	v_R	rated wind speed
MDP	Markov decision process	x^{SPV}	number of solar PV panels installed
		x^{W}	number of wind turbines
		x^{ICG}	size of the ICG

A number of recent studies have focused on optimum design of distributed energy systems for multiple energy services considering both grid connected and standalone operation using bi-level optimization algorithms [7,8]. In these algorithms, the operation strategy (dispatch) is optimized at the secondary level, which is used to compute the operation cost of the system considering hourly changes in demand and renewable energy potential and grid conditions [9–11]. The approach has been extended later to optimize multiple energy systems [12,13]. Primary level computation is used to optimize the selection of energy system components based on the dispatch strategy optimized at the primary level. Bi-level optimization algorithms require higher computational time especially in the context of considering non-dispatchable renewable energy technologies. For example, the bi-level optimization algorithm developed by Evins [11] took approximately three days to optimize to design distributed multi-energy hubs using parallel computation (operation of the energy system is optimized considering 8760 time steps using MILP and system design using heuristic algorithms). The higher computational time to formulate objective functions when using Actual Engineering Models (AEM) is considered as one major limitation. Promising and computationally efficient methods are required to evaluate the impact of uncertainties during the optimization of the energy system design [14].

Recently, surrogate models or meta-models have been used in a number of different fields to reduce the computational time of optimization problems. A surrogate model is used to bypass the AEM, which takes more computational time when mapping decision space variables into objective space. Black-box methods such as neural networks, support vector machines (SVM), random forests etc. have been used to develop surrogate models. According to recent literature, surrogate models have shown the potential to speed up computation more than 100 times while maintaining an up to 90% higher accuracy [15]. This has made surrogate models popular in a wide area of engineering fields such as civil, automobile, aerospace and manufacturing engineering. Surrogate models have been used frequently for building simulation in order to reduce the simulation time in building energy optimization problems [16,17]. Furthermore, components of energy systems have been optimized using surrogate models. For example, Miao et-al. [18] and Cheng et-al. [19] designed the PEM fuel cell using a surrogate model while Thomsen et-al. [20] and Halder et-al. [21,22] used surrogate models to optimize the design of wave energy converters. Kadhim and Rona [23,24] optimized the design of an axial turbine used in liquefied natural gas plants. Most of these studies used surrogate models to replace time consuming computational fluid dynamic (CFD) models. However, moving from a single component to an energy system

entails a number of complexities.

Surrogate models have been seldom used to support energy system optimization. Bornatico et-al [25] use surrogate model to optimize collector area and storage size of a solar thermal energy system. Simply two variables related to the design are considered in this study while surrogate model is directly coupled with the heuristic optimization algorithm to obtain Pareto fronts considering multiple objectives. Sanchez and Martin used a surrogate models to presents the complex formulation of chemical plant operation when optimizing the design of an Ammonia production using renewable energy technologies [26]. Similarly to Bornatico et-al. [25], the operation strategy of the energy system is not considered when developing the surrogate model.

None of the previous studies have considered the dispatch strategy when developing the surrogate model. Considering the dispatch strategy will extend the number of decision variables significantly and make the surrogate model to be complicated. As a result, using surrogate model alone may lead to sub-optimal design solutions due to the limitations in approximating. Hence, it is important to look into promising methods that can be used to combine surrogate models with AEMs during the optimization process which has not been considered in the present state of the art from the energy system design perspective. In addition to these limitations, bi-level often requires to restart from the beginning whenever there is a reasonable change in the renewable energy potential, demand and techno-economic data increasing the computational time notably. This limits the applicability of such optimization models in regional and national scale where large number of distributed energy systems need to be optimized. None of these limitations have been addressed in the present state of the art.

Towards addressing the limitations, the design process of an electrical hub operating in connection to the grid is considered in this study. Electrical hub consists of wind turbines, SPV panels, battery bank and an internal combustion generator (ICG). A computation platform is developed by combining a surrogate model with an optimization algorithm to assist energy system optimization. To overcome the above stated limitations, the following objectives are expected to be met in this study:

- Both system design and dispatch strategy are considered (besides being limited to system design only) in the surrogate-model and optimization process. A comprehensive study is conducted to obtain the best fitting surrogate model after considering eight configurations.
- A novel computational algorithm is developed to assist the optimization process which combines both surrogate and actual

engineering model.

- Since the supervised learning model cannot be used directly when moving into other cases (where the renewable energy potential, energy demand and techno-economic data might be different from the initial case that the supervised learning model is trained for), a transfer learning technique is proposed to adapt the surrogate model. Such methods can significantly reduce the computational time when analyzing energy systems at the regional or national scales.

The current research article is arranged as follow. [Section 2](#) presents an outline of the computational model proposed in this study. [Section 3](#) presents the actual engineering model (AEM) used to develop the surrogate model as well as to optimize the energy system. [Section 4](#) describes the supervised and transfer learning techniques used when developing the surrogate model. [Section 5](#) presents the novel optimization algorithm developed in this work. Applicability of the model considering different cases and future applications of the model are investigated in [Section 6](#). Finally, conclusions are discussed in [Section 7](#).

2. Overview of the computational model

Energy system designing processes are often conducted as a simulation based optimization problem. The response of the energy system to the varying renewable energy potential, demand and grid conditions should be assessed when mapping decision space variables into the objective space. Usually, a time series simulation of 8760 time steps (24×365) or a set of representative time steps are considered during the simulation which can be considered as a Markov Decision Process (MDP). The objective function values depend on both system configuration and operation strategy. Hence, optimizing both system configuration and dispatch strategy while going through the time series simulation usually takes more computational time (in certain instances up to several days). This section presents an outline of the computational algorithms used and the proposed novel frame work based on supervised and transfer learning methods. [Section 2.1](#) explains the present approaches while [Sections 2.2](#) and [2.3](#) present newly introduced computational models that can be used to reduce computational time when designing distributed energy systems. [Section 2.4](#) presents an approach that can be used to generalize the computational model for other locations with the support of a transfer learning technique.

2.1. Actual engineering model (AEMs)

Energy system operation (selection of dispatchable source, storage and grid) has a direct influence on optimum system sizing. Therefore, operation of the energy system should be considered on an hourly basis

to take into account the changes in renewable energy potential, demand and grid conditions. According to [\[27,28\]](#), two approaches are used in this context, aligned with dynamic programming. The first method is closely related to the value function method where operation strategy of the energy system is optimized on an hourly basis. This is usually practiced when designing a complex poly-generation system, which will result in extensive computational time. The second method is based on a policy function method where it is assumed that there is an optimum policy that governs the operating state of the energy system, which has to be optimized along with the system configuration. The operation of the energy system needs to be considered on the hourly temporal scale throughout the years in both the scenarios, which will result in a higher computational time ([Fig. 1](#)). Hence, irrespective of the adopted method, the energy system optimization process takes longer. This study uses the second approach for AEM, which the model is used to obtain a training data set for the surrogate model. Furthermore, AEM is used in the proposed novel computational algorithm to map the decision space variables into the objective space with the support of the developed surrogate model. A comprehensive overview of the AEM is presented in [Section 3](#).

2.2. Supervised learning to develop surrogate model

Surrogate models are black box models such as neural network, Support Vector Machines (SVM) etc. They simply present the relationship between input and output while bypassing the use of computationally extensive models ([Fig. 2](#)). Although black box models are fast in computation, they cannot completely replace AEMs. Data driven approaches such as surrogate models can minimize the extensive computational processes and detailed information related to both analytical and numerical models. Specific to the energy system optimization process, surrogate models are used to reduce the computational time required for time series simulation when mapping decision space variables to the objective space. Three steps are required when using the surrogate models to design energy systems:

- (1) Create a data set for training and validating the surrogate model.
- (2) Use a supervised learning method to develop the surrogate model.
- (3) Optimize the energy system using the surrogate model.

The AEM described in [Section 2.1](#) is used to create the data set that is used to create the surrogate model. Subsequently, several supervised learning techniques have been used to train the surrogate model and evaluate its performance. A comprehensive overview of these techniques can be found in [Section 4](#). The surrogate model only maps the decision space variables into the objective space. In order to optimize the design, an optimization algorithm needs to be used to obtain an optimum design by using the surrogate model.

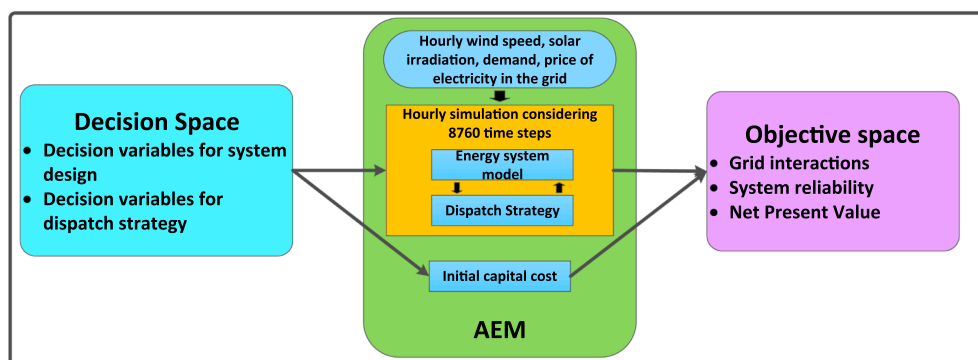


Fig. 1. Flow chart for the Actual Engineering Model (AEM). Decision space variables are mapped into the objective space through the AEM. A detailed description about the AEM is presented in [Section 3](#).

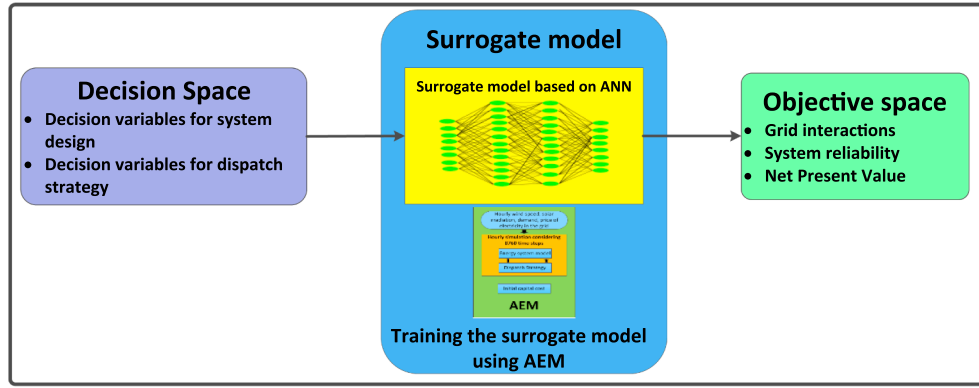


Fig. 2. Flow chart for the surrogate model. Decision space variables are mapped into the objective space through the surrogate model. AEM is used to generate a data set in order to train the surrogate model. A comprehensive overview of the surrogate model developed in this study is presented in [Section 4](#).

2.3. Hybrid optimization algorithm combining surrogate and AEM models

AEMs are more accurate in comparison to surrogate models when computing the objective function values. However, it takes much longer to compute the objective function values using AEM. Furthermore, developing a surrogate model with a very high level of accuracy will demand extensive training of the surrogate model and this in turn will demand a larger set of training data resulting in more computational time. An optimum combination of a surrogate model and an AEM will help to sort out this issue while significantly minimizing the computational time required for the energy system design process. The Hybrid Optimization Algorithm (HOA) combines both AEM and surrogate models in order to speed up the computation ([Fig. 3](#)). In the HOA, the surrogate model is initially used with the optimization algorithm to generate the initial Pareto front. Thereafter, the initial Pareto front is used as the starting point for the secondary stage optimization which is using AEM. The higher computational speed of the surrogate model helps the optimization algorithm to reach the Pareto front faster. However, AEM is used in the second stage which maps the decision space variables into the objective space with a better accuracy. In HOA, the surrogate model accelerates reaching the actual Pareto front, which will be refined using the AEM (which is described in detail in [Section 3](#)).

2.4. Transfer supervised learning algorithm

The development presented in [Sections 2.2 and 2.3](#) are focused on optimizing distributed energy systems for a specific location. Often, it is required to start the optimization process from the beginning when moving to another location. This is one major limitation which hinders the use of AEM at the national or regional scale where a large number of distributed energy systems need to be optimized considering their interactions. One major advantage of a data driven approach is that the model obtained using supervised learning can be adapted to other locations with certain changes. The adapting process is not as intensive as the initial training process ([Fig. 4](#)). As a result, the computational time required to develop the surrogate model is significantly reduced. The process used to adapt the surrogate model is known as transfer learning. The main advantage of transfer learning is that it minimizes the amount of labeled data required for the training process while minimizing the time required to generate the labeled data and the training process. [Fig. 4](#) presents the approach used in the present study for the model adaptation. The surrogate model initially developed for a specific location is adapted for another location using transfer learning. A set of labeled data is generated for the new location using the AEM for the transfer learning process. Accelerated GPU (Graphics Processing Unit) computing is used to speed up the process of generating data. By using the labelled data generated for the new location, the surrogate model is transfer learned. Finally, the Surrogate Model trained using Transfer

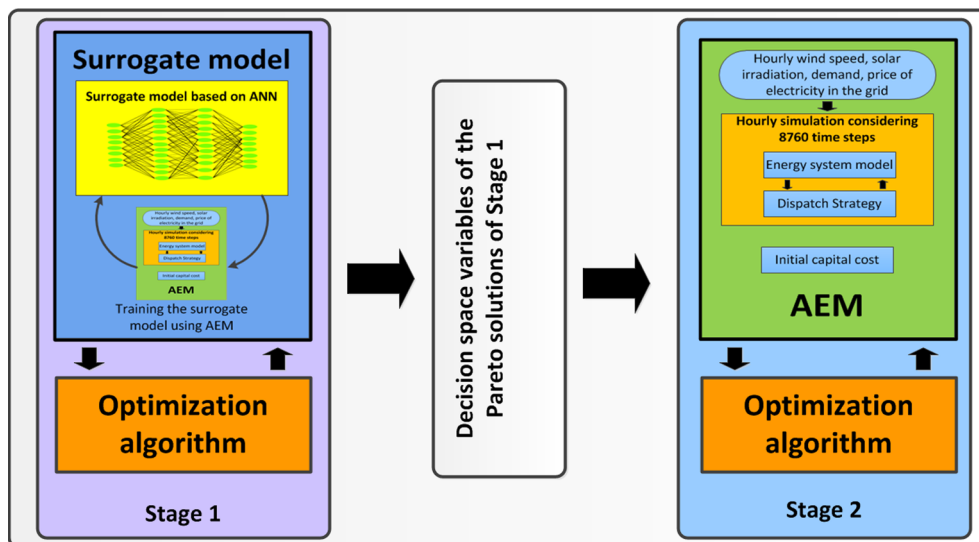


Fig. 3. Flow chart for HOA combining the surrogate model and AEM. The surrogate model helps to come up with a better starting point to the AEM, thus speeding up the computation process. A comprehensive overview of the HOA is presented in [Section 5](#).

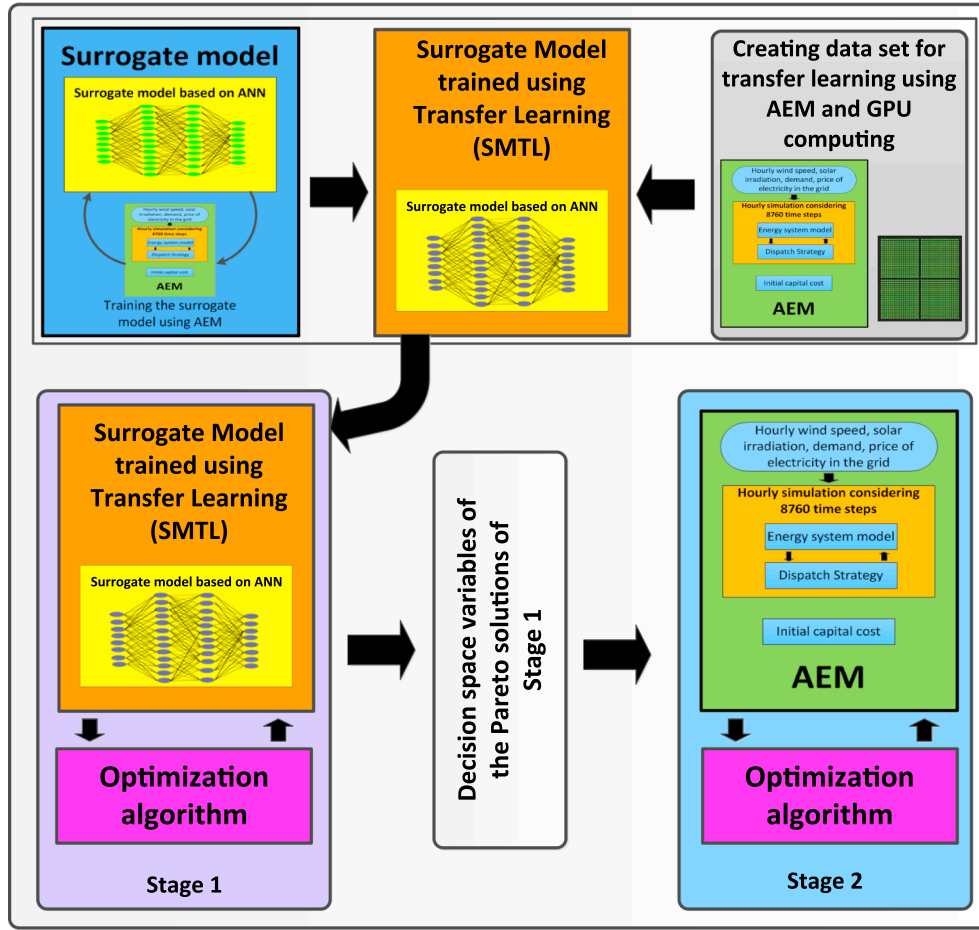


Fig. 4. Outline of the architecture used to develop SMTL and combine the SMTL with the HOA.

Learning (SMTL) replaces the surrogate model trained using supervised learning in the HOA as shown in Fig. 4.

3. Mapping of decision space variables to the objective space using AEM

An energy hub consisting of renewable energy sources, energy storage and ICG connected to the grid is considered in this study when developing the AEM. Techno-economic aspects related to the energy system are considered in the computational model using lifecycle simulation. The computational model evaluates the lifecycle cost, system autonomy, reliability, utilization of renewable energy, etc. through a life cycle simulation. The AEM presents the simulation based computational model used to map decision space variables representing both system design and operation strategy into the objective space.

3.1. Energy flow model

The energy flow of the system is simulated on an hourly basis. Time series data for wind speed, solar irradiation and ambient temperature are taken from a meteorological database. Hambantota, a southern coastal city in Sri Lanka with high wind and solar energy potential is considered for the case study.

3.1.1. Solar power generation

Power generation from the solar PV panels depends on several factors such as global solar irradiation on the solar PV panel, efficiency of the solar PV panel and the number of PV panels installed. Hourly global solar irradiation data on a horizontal surface is obtained from the

meteorological records. Climed and Klucher models are used to compute tilted global solar irradiation based on the global horizontal solar irradiation. A detailed description of the solar energy model can be found in [29]. The Duricsh model [30] is used to compute the efficiency of the PV panels. The Duricsh model [30] considers the impact of cell temperature (θ), air mass (AM), global solar irradiation (G_t^β) and the type of solar PV panel when computing the efficiency of the solar PV panels (Eq. (1)).

$$\eta_t^{SPV} = p^{SPV} \left[q^{SPV} \left(\frac{G_t^\beta}{G_0^\beta} \right) + \left(\frac{G_t^\beta}{G_0^\beta} \right)^{m^{SPV}} \right] \left[1 + r^{SPV} \left(\frac{\theta_t^{SPV}}{\theta_0^{SPV}} \right) + s^{SPV} \left(\frac{AM}{AM_0} \right) + \left(\frac{AM}{AM_0} \right)^{u^{SPV}} \right], \quad \forall t \in T \quad (1)$$

Standard values for G_0^β , θ_0^{SPV} , AM_0 are taken respectively as $G_0^\beta = 1000 \text{ W m}^{-2}$, $\theta_0^{SPV} = 25^\circ\text{C}$ and $AM_0 = 1.5$. Parameter values of p^{SPV} , q^{SPV} , r^{SPV} , s^{SPV} , m^{SPV} , u^{SPV} for different SPV technologies, such as mono-crystalline, polycrystalline and amorphous silicon cells, are taken from Ref. [31]. The number of solar PV panels installed (x^{SPV} ($x^{SPV} \in \mathbb{N}$)) determines the overall power generation from the PV system which is considered as a decision space variable (also an input to the surrogate model). The hourly power supply from the SPV panels P_t^{SPV} is calculated according to Eq. (2). A^{SPV} and represents the area of a single SPV panel as well as the number of SPV panels.

$$P_t^{SPV} = G_t^\beta \eta_t^{SPV} A^{SPV} x^{SPV}, \quad \forall t \in T \quad (2)$$

3.1.2. Wind power generation

$$\tilde{P}_t^w = \begin{cases} \tilde{P}_t^w = 0, & v_{CI} > v_t \\ \tilde{P}_t^w = a_1^w v_t^3 + b_1^w v_t^2 + c_1^w v_t + d_1^w, & v_{CI} < v_t < v_1 \\ \tilde{P}_t^w = a_2^w v_t^3 + b_2^w v_t^2 + c_2^w v_t + d_2^w, & v_1 < v_t < v_2 \\ \vdots & \vdots \\ \tilde{P}_t^w = a_{n_s}^w v_t^3 + b_{n_s}^w v_t^2 + c_{n_s}^w v_t + d_{n_s}^w, & v_{n_s-1} < v_t < v_{n_s} (v_R) \\ \tilde{P}_t^w = P_R, & v_R < v_t < v_{CO} \\ \tilde{P}_t^w = 0, & v_t > v_{CO} \end{cases}, \quad \forall t \in T \quad (3)$$

Similar to the power generation from solar PV panels, power from a single wind turbine is computed using cubic spline interpolation method [32,33]. The Cubic Spline Interpolation method splits the power curve of the wind turbine (given by the manufacturer) into $(n_s + 1)$ points and subsequently interpolated using a 3rd order polynomial function according to Eq. (3).

In Eq. (3), a_i^w , b_i^w , c_i^w , and d_i^w are coefficients of the polynomial function which vary depending on the 'power curve'. v_R , v_{CI} , v_{CO} and P_R denote rated wind speed, cut-in wind speed, cut-off wind speed and rated power of the wind turbine. The net power generation from wind turbines can be computed by taking power generation of a single wind turbine multiplied by the number of wind turbines. The number of wind turbines (x^w ($x^w \in N$)) in the design is a decision space variable considered in the optimization.

$$P_t^W = \tilde{P}_t^W(v_t) x^w \eta^{W-losses}, \quad \forall t \in T \quad (4)$$

$\eta^{W-losses}$ accounts for other losses that take place in the energy conversion.

3.2. Modeling dispatchable energy sources and dispatch strategy

The energy system consists of a battery bank and an ICG which work as dispatchable energy storage and source respectively. The model considers operating constraints of system components. Maximum power flow from the battery bank is determined based on the SOC of the battery bank and maximum power flow of the ICG is determined based on the nominal power. Size of the ICG is considered as a decision space variable (x^{ICG} ($x^{ICG} \in N$)). Operating load factor of the ICG is determined using the dispatch strategy. Similar to the ICG, capacity of the battery bank is considered as a decision space variable (x^{Bat} ($x^{Bat} \in N$)). A constant round cycle efficiency of 85% is considered for charging and discharging processes for the battery bank.

State of Charge (SOC) model is used to compute the charge levels of both battery bank and thermal storage. SOC of the battery is calculated according to Eq. (5)

$$SOC_{t+1} = SOC_t(1 - \sigma_t) + I_{batt} \Delta t \cdot \eta_{ch}/C_{bat} \quad (5)$$

In Eq. (5), σ denotes self-discharge coefficient, which was taken as 0.02%, η_{ch} and C_{bat} denotes the round cycle efficiency of the battery bank and its capacity. A similar approach is used for the thermal storage.

A bi-level dispatch strategy is used in this study for the energy management of the power system. A comprehensive description of the dispatch strategy is presented. Dispatch strategy determines the operating load factor of the ICG, interactions with the battery bank and grid. The primary level of the dispatch strategy is based on fuzzy automata. The primary level is used to determine the operating load factor of the ICG. Depth of discharge of the battery bank (y_1), load mismatch between demand and renewable energy generation (y_2) are considered as the inputs to the primary level of the dispatch strategy. Set of weights (w_i ($w_i \in N$)) corresponding to the fuzzy rules are considered as decision space variables which are optimized subsequently. Finite state automata theory is used for the secondary level which determines the

interactions with the grid and energy storage. Power generation within the system, energy demand, state of charge of the battery bank and grid curtailments are considered at the secondary level when interacting with the grid energy storage. The secondary level operates with a finite set of states (s ($s \in S$)) where the state transfer rules (r ($r \in N$)) are optimized using the optimization algorithm. An extended explanation of the dispatch strategy can be found in [28,34].

3.3. Formulation of objective functions and constraints

The set of decision spaces described in Sections 3.1 and 3.2 is mapped into the objective space through a time series simulation to form the objective functions. Two objective functions are considered in this study for the Pareto optimization along with a set of constraints. Net present value of the system (NPV o_1 ($o_1 \in O$)) and grid integration level o_2 ($o_2 \in O$) are considered as the objective functions. The level of power supply reliability c_1 ($c_1 \in C$) is introduced as a constraint to the optimization algorithm (other constraints related to the limitations in power conversion devices, grid curtailments etc. are handled in the dispatch strategy).

3.3.1. Power supply reliability

Maintaining power supply reliability is considered as a major priority. Therefore, power supply reliability is considered as a constraint in the optimization process. Power supply reliability has been amply used in recent literature as a performance indicator (as an objective function and a constraint) [28,35]. Loss of power supply will take place whenever the electricity demand is greater than the electricity demand of the time step ($\forall t \in T$). Based on that loss of load probability (LOLP (c_1 ($c_1 \in C$))) is computed according to Eq. (6).

$$LOLP = \frac{\sum_{t \in T} ELD_t - P_t^{SPV} - P_t^{gen} - P_t^{Bat, max} - P_t^{FG max}}{\sum_{t \in T} ELD_t}, \quad (c_1 \in C) \quad (6)$$

In this equation, ELD_t , P_t^{gen} , $P_t^{Bat, max}$, $P_t^{FG max}$ respectively present electricity load demand, power generation capacity (maximum) and maximum power that can be taken from the battery bank and grid.

3.3.2. Formulating net present value (NPV)

The Net present value (NPV) of the energy system has been amply used as an objective function in power system optimization. It presents the financial aspect of the project which is often given a high priority. In this study, NPV is formulated (according to Eq. (7)) considering initial capital cost (ICC), fixed operating and maintenance (OMF) and variable operating and maintenance cost (OMV). ICC includes all the cash flows representing acquisition and installation costs of system components such as wind turbines, solar PV panels, battery bank, ICG, inverters etc. Initial capital cost value for system components are taken from Ref. [28,28,36].

$$NPV = ICC + \sum_{s \in S} (OMF_s CRF_s) + \sum_{l=1}^H \sum_{s \in S} p^l OMV_{s,k} + CRF_G \sum_{t \in T} (PTG_t GCF_t + PTG_t GCT_t), \quad (o_1 \in O) \quad (7)$$

In this equation, CRF, p , PTG, GCF and GCT denote capital recovery factor, real interest rate, power units sell back to the grid, price of grid electricity when purchasing electricity and selling back respectively.

3.3.3. Presenting grid integration level (GI)

The importance of energy autonomy has been widely discussed with the emergence of net-zero energy buildings and neighborhoods. Grid integration level has been amply used as an objective function which

Represent the autonomy level of the energy system. According to Perera et-al, grid integration can be defined in different ways [28,37]. Being align with Ref. [38,39], defines grid integration level according

to Eq. (8).

$$G_{FG} = \sum_{\forall t \in T} P_t^{FG} / \sum_{\forall t \in T} P_t^{ELD}, \quad (o_2 \in O) \quad (8)$$

In this equation, P_t^{FG} denotes the power units purchased of the grid.

4. Hybrid surrogate model to present energy system

There are number of different methods such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), Linear Regression (LR) etc. that can be used to develop surrogate models. In recent past, ANN has shown to be promising in number of areas with the arrival of deep learning. Therefore, ANN are selected to be used for the surrogate model in this study. In simple words, ANN consist of set of neurons where each neuron facilitates a non-linear mapping between input and output. However, arriving at the optimum architecture for the neural network is a difficult task due to the larger number of combinations that are possible to consider. Hence, a pool of combinations is considered for the architecture which consist of neural networks having different number of layers and neurons per each layer as shown in Table 1. The architectures considered in this study consist of neural networks consisting of two layers up to thirteen while increasing the complexity of the architecture.

Optimization is done using Levenberg-Marquardt backpropagation (damped least square) method. Mean Square Error (MSE) is used as the loss function and no regularization term is used. Training, validating and testing datasets used in the training of initial model consist of 448,000, 192,000 and 640,000 respectively. Afterwards, transfer learning is used to adapt the initial neural network to other scenarios. Transfer learning is a strategy in machine learning used to adopt a model trained on one dataset to another which is not exactly same, but is related. In this case, it is done by continuing the training of the initial model on a dataset corresponding to the scenario that needs to be adopted. This dataset is smaller in size compared to the dataset used in the initial training. It consists of 7000 samples for training, 3000 samples for model validation and 10,000 for testing. During the transfer learning stage, the same parameters are used as in the initial training stage apart from the difference in dataset.

The performances of these neural networks are evaluated based on Mean Absolute Error (MAE). Based on this metric, it is observed that increasing the number of layers as well as neurons per layer decrease the MAE. However, for the architectures that are tested, best performance is obtained with a shallow network with 2 layers and each of them consisting of 50 neurons. Results for y1 (output Variable 1 (O1)) are summarized in Table 1. When analyzing the results, it is prudent that accuracy of the prediction increases by increasing the number of layers. This can be observed when moving from AR1 to AR6 where the mean AEP reduces from 4.35 to 2.85. However, extending the ANN by adding layers will demand for higher computational time for training while demanding more computational time during the prediction. Hence, the possibility to improve the prediction error by widening the neural network is subsequently considered. AR7, 8 and 9 are introduced by widening the neural network. When moving from AR6 to AR7 a significant drop in the Mean AEP is observed. However, an improvement in Mean AEP is observed when moving from AR6 to AR8 with the significant widening of the neural network architecture even with two hidden layers. To analyze the influence of both widening and deepening, two more hidden layers are added to AR8 and created AR 9. A significant improvement in Mean AEP is observed as a result of the modification. However, the modification introduced in AR9 increases the number of parameters as well as the prediction time notably. As a result, AR9 cannot be used although having a higher accuracy. AR8 is used in this study as the architecture for the surrogate model.

5. Optimization algorithm

Optimization algorithms have been used in two different parts of this study. An optimization algorithm is used initially to train the surrogate model. Afterwards, it is used to optimize the energy system. The stochastic gradient decent (SGD) algorithm is used to train the surrogate model. A concise overview of SGD is presented in Section 4. This section is focused on the second application where optimization of the energy system is considered.

Both AEM and surrogate model map the decision space variables into the objective space. AEM uses a time series simulation where both system configuration and policy function related to the dispatch strategy are considered as the decision space variables (Table 2). The surrogate model uses a fully connected feed forward neural network to map the decision space variables into the objective space. An optimization algorithm is required to arrive at the optimum solution irrespective of the method used to map the decision space variables. When it comes to energy system optimization, there are a number of instances where the objective functions have been formulated considering the restrictions in the optimization algorithm. As a result, linear programming and mixed integer linear programming have been amply used when designing distributed energy systems. At the same time, a number of recent studies have used heuristic methods to design distributed energy systems. However, the objective functions obtained using the surrogate models based on neural networks are neither linear nor convex. The optimization algorithm introduced in this study should possess the capability to interact with both AEM and the surrogate model. Hence, it is inevitable to consider an optimization technique other than a heuristic method.

5.1. Implementation of the optimization algorithm along with AEM and surrogate model

The optimization algorithm is connected with the surrogate model and AEM according to Fig. 5. Two orange blocks represent the elements of the optimization algorithm. The light blue and green blocks represent the elements of the surrogate model and AEM respectively. The steady ϵ -State Evolutionary Algorithm [40] is used for the optimization algorithm which is proven as a method to maintain the diversity while reaching the final set of Pareto solutions. The polynomial mutation operator [41] and simulated binary crossover operator [42] are used along with differential evolutionary operators [43–45] in the reproduction of the population. The Constraint Tournament Method [41] is used to consider the constraints that are not handled at the level of dispatch strategy. Computational time for the Pareto front depends on the objective functions selected and the number of generations considered.

The optimization algorithm follows Path 1 (orange-blue-orange blocks) when optimizing the energy system using the surrogate model. The decision vector required to evaluate is transferred to the blue box.

Table 1
Architecture of the neural networks considered in this study.

Neural Network	Number of hidden layers	Number of neurons in each layer	Mean AEP
AR1	2	16, 8	4.35
AR2	3	18, 14, 8	3.39
AR3	5	18, 18, 14, 10, 6	2.79
AR4	5	20, 18, 18, 16, 16	2.80
AR5	9	20, 18, 18, 18, 18, 16, 16, 16, 16	2.93
AR6	13	20, 18, 18, 18, 18, 18, 18, 16, 16, 16, 16, 16, 16	2.85
AR7	2	25, 25	3.09
AR8	2	50, 50	2.33
AR9	4	50, 50, 50, 50	1.48

Table 2

Decision space variables considered for the optimization problem including both system configuration and operation strategy.

Variable	Lower bound	Upper bound	Interval	Description
SPV Type (N^{TY-SPV})	0	3	1	Mono-crystalline, Polycrystalline and Amorphous ¹
# SPV Panels N_{SPV}	0	120	1	
Type of Turbines (N^{TY-W})	0	2	1	1, 5 kW
# Wind Turbines	0	30	1	
# Battery banks	0	20	1	0–240 ² kWh
ICG Capacity (kVA)	0	15	0.5	
W_{ij} (weight matrix)	0%	100%	Continuous	
Seven rules (r ($r \in N$)) corresponding to the finite states			Continuous	

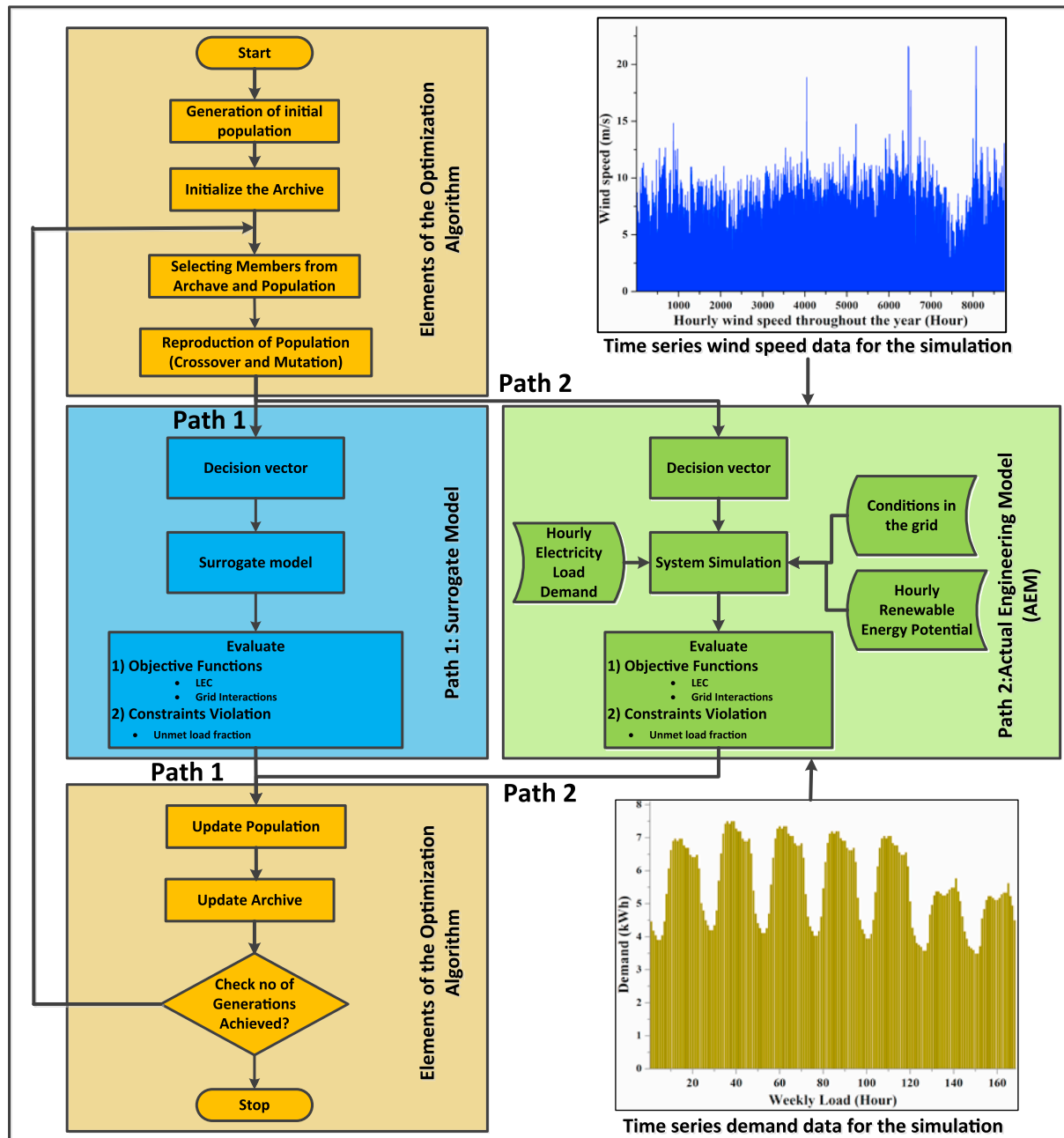
¹ 0.5 kW maximum capacity.² Each battery bank having 12 kWh capacity.

Fig. 5. Presents the implementation of the optimization for the surrogate model and AEM. The two orange boxes represent the elements of the optimization algorithm common for both the surrogate model and AEM. Path 1 is taken when using the surrogate model (blue box) while Path 2 is taken when using the AEM (green box). LEC and grid integration levels are used as the objective functions. Unmet load fraction is considered as a constraint. A detailed description about the computation of LEC is presented in [Section 3](#).

The surrogate model directly maps the decision vector into the objective space which helps to obtain the objective function values and constraint violation. These values are transferred to the orange block, which evaluates the objective function values and subsequently updates the population and archive. Similar to the surrogate model, Path 2 is followed when using the AEM (orange-green-orange blocks). The decision vector required to evaluate is transferred to the green block. As opposed to Path 1, a detailed time-series simulation is conducted to map the decision vector into the objective space. Hourly wind speed, solar irradiation and energy demand data are used for the time series simulation as explained in Section 3. Subsequently, objective function values and constraint violation are transferred to the orange block to update the population and archive.

5.2. Hybrid optimization algorithm (HOA) combining AEM and surrogate model

Section 5.1 describes the outline of the optimization algorithm when using either the surrogate model or AEM. However, the same structure cannot be adapted when combining these two algorithms. The surrogate model is faster when computing the objective function values. However, the accuracy of the objective functions and constraints might not be very high in certain instances. In contrast, AEM takes longer for

the computation but shows higher accuracy when computing the objective functions and constraints. Hence, the surrogate model is used to arrive at a better starting point to the optimization algorithm based on AEM. As shown in Fig. 6, an optimization algorithm based on the surrogate model is used as the first step. Subsequently, the population and archive (Pareto solutions) are moved to the next step. However, it is important to check the values for the objective functions and constraints before direct use in the AEM (since there are certain limitations in the surrogate model when computing both objective functions and constraints). Therefore, in Step 2, AEM is used to simulate both archive and population (obtained using the optimization algorithm based on the surrogate model). Afterwards, dominance of the Pareto solutions is rechecked and, population and archive (set of Pareto solutions) are updated. The updated archive and population are transferred to Step 3. Updated archive and population obtained from the surrogate model are an efficient starting point for the optimization algorithm. Thereafter, the optimization algorithm will use the AEM in order to further improve the Pareto solutions using AEM.

6. Results and discussion

This section is divided into two main parts. The first part demonstrates the development of supervised and transfer learning models

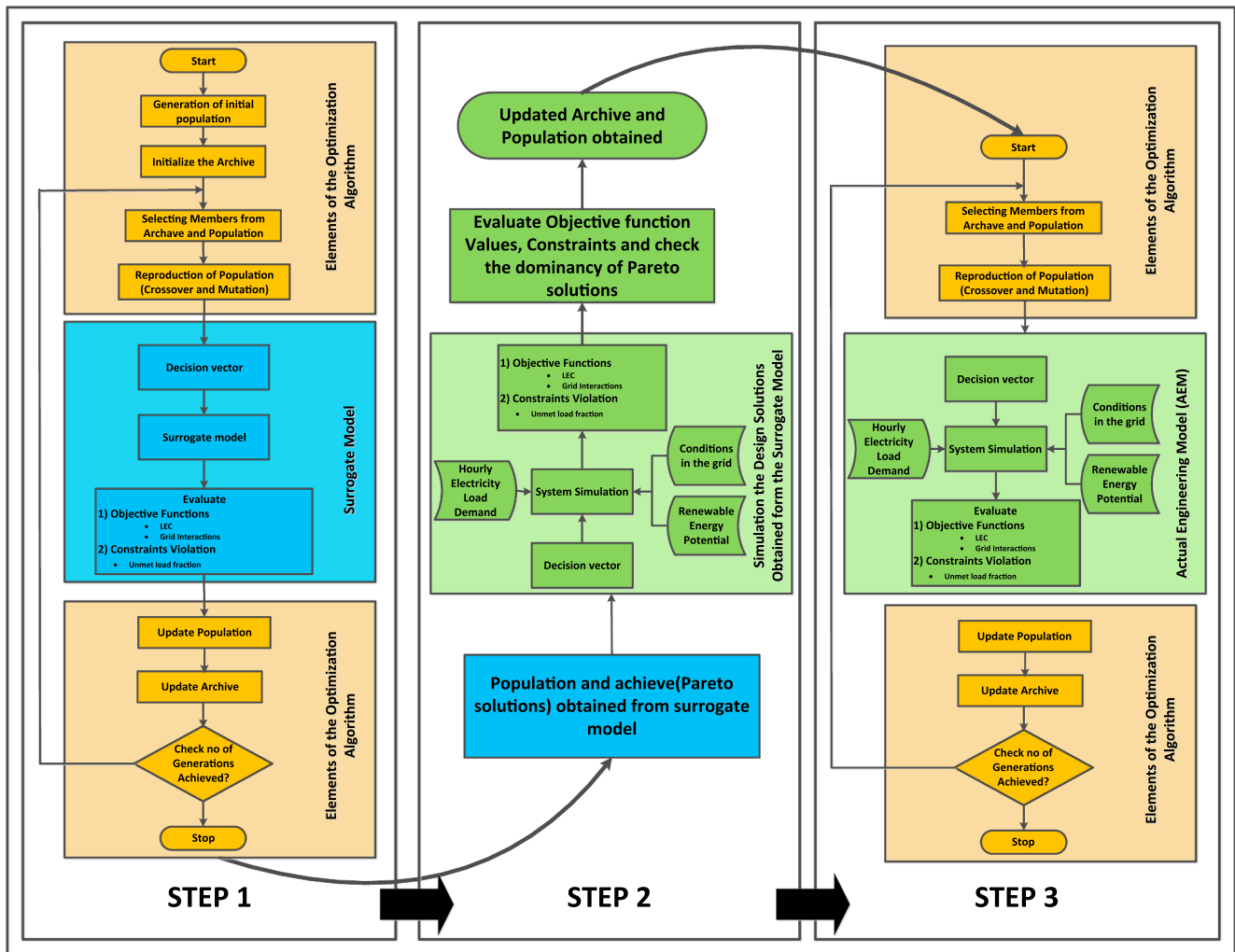


Fig. 6. Presents the implementation of the Hybrid Optimization Algorithm (HOA), which consists of three steps. A surrogate model is used initially to reach a better starting point with regard to the AEM in a short period of time (Step 1). In Step 2, Pareto solutions obtained using the surrogate model are simulated again using the AEM and the Pareto solutions are updated based on the objective function and constraint values obtained using the AEM. Subsequently, these Pareto solutions are transferred into the optimization algorithm which uses AEM in Step 3 in order to obtain the final Pareto solutions.

used for the replication of the AEM. The second part is devoted to analyzing the performance of different techniques that are used to combine AEM with surrogate models that are developed using supervised and transfer learning techniques.

6.1. Comparison of the Pareto fronts obtained using AEM and surrogate models

Pareto fronts obtained considering both AEM and surrogate models are presented in Fig. 7. The Pareto fronts overlap when the grid integration levels are higher (Region Q in Fig. 7). However, they start to divert from each other when grid integration levels are below 10% (Region P in Fig. 7). NPV begins to increase at a much higher rate for the Pareto front obtained using the surrogate model when the grid integration levels are below 10%. As a result, a significant difference in NPV can be seen when the grid integration levels are within the range of 1–5%. The difference in NPV between AEM and the surrogate model slightly decreases when reaching the fully autonomous state (due to the sudden increase in NPV observed in AEM Pareto front when reaching the fully autonomous state). The Pareto front only indicates the reflection of the decision space variables on the objective space. The most important factor to be analyzed is the similarity between the decision space variables corresponding to Pareto solutions which are obtained using two models that are close to each other.

In order to understand the applicability of the surrogate model, Pareto solutions with similar grid integration values are taken from both Pareto fronts and tabulated in Tables 3 and 4. Table 3 contains the design solutions which maintain grid interaction levels above 10% while Table 4 presents the design solutions with grid interaction levels less than 10%. Comparing the design solutions is challenging due to the higher dimensionality of the decision space. Besides taking all the decision space variables, system configuration variables are tabulated in both Tables 3 and 4. The ratio between solar PV to wind turbine installed capacity and dispatchable to non-dispatchable installed capacity is taken to support the analysis. Designs A, B, C and D respectively present design solutions with similar grid interaction levels. Respectively, A-AEM and A-S are obtained using actual engineering and surrogate models.

Table 3 presents the Pareto solutions from the section where both AEM and surrogate models overlap (with grid integration equal or above 10%). When comparing the Pareto solutions, it is clear that the design solutions look close to each other when comparing the wind turbine, solar PV, battery bank and ICG capacities. When comparing the two sets, it can be seen that the solar PV capacity is slightly higher in the Pareto solutions obtained using AEM. For example, installed solar PV capacity decreases by respectively 7.8, 3.9 and 6.5 kVA when moving from AEM to surrogate models (respectively in A, B, C), resulting in a change of 11.5, 5.7 and 10.4% respectively. In contrast, wind turbine capacities are higher in Pareto solutions obtained using the surrogate model. For example, wind turbine capacity increase by 10, 20 and 15 kVA when moving from AEM to surrogate models (respectively in A, B, C), resulting in a change of 13, 29 and 15% respectively. When compared to solar PV capacity, wind turbine capacity shows a significant deviation. As a result, the ratio of wind turbine capacities to solar PV capacities increases when moving from the AEM to the surrogate model. Similarly, a deviation in ICG and battery bank capacities can be observed. Finally, it can be concluded that the system configuration shows notable deviations although the objective function values look quite closer each other when the grid integration level is above 10%.

When moving from Tables 3 and 4, a clear deviation in the objective function values can be noticed. A significant increase in NPV is observed when moving from design solutions obtained using AEM to the surrogate model (as observed in Fig. 7). This makes it interesting to analyze the design solutions further in order to reason out the observed deviations. When comparing the design solutions of the two sets, a

slight deviation in solar PV panel capacity is observed. Major deviations can be observed in the two sets when considering wind turbine, ICG and battery bank capacities. Wind turbine capacity drops when reducing the grid interaction levels for the Pareto solutions of the surrogate model. For example, wind turbine capacity drops from 65 kVA to zero when moving from E-S to I-S. As a consequence of the reduction in wind turbine capacity, the ratio of wind turbine to solar PV capacity gradually drops in Pareto solutions obtained using the Surrogate model when reducing the grid interactions. In contrast, the capacities of both battery bank and ICG increase with the reduction in grid interactions. For example, the ICG capacity increases from 10 to 25 kVA. As a consequence of the reduction in wind turbine capacity and increase in the ICG capacity, the ratio between dispatchable to non-dispatchable capacities increases significantly when minimizing grid interaction levels for the Pareto solutions obtained using the surrogate model. Such clear patterns in system configuration are not visible for the Pareto front solutions obtained using the AEM although complex patterns can be observed for the decision space variables corresponding to the dispatch strategy.

The patterns observed in the Pareto solutions obtained using the surrogate model such as increase in energy storage, dispatchable energy source and reduction in non-dispatchable energy capacity (wind turbines since seasonal fluctuations in wind speed are significant compared to solar energy potential) are typical conditions expected in standalone operation [29,46]. However, introducing grid integration and advanced energy management enables the system to maintain a higher autonomy without a significant change in the energy system for the specific example. As a result, a noticeable pattern in system configuration is not observed for the Pareto solutions obtained using the AEM. When comparing the design space variables, it is evident that the complexity introduced by the decision space variables corresponding to the dispatch strategy has not been properly learnt by the surrogate model using the supervised learning method when grid integration levels are low. As a consequence, the Pareto front obtained using the surrogate model diverts from the one obtained using AEM following the general trend. However, the surrogate model presents better results when it comes to higher grid integration scenarios. This makes it important to further train the surrogate model considering scenarios where grid integration levels are low. Active learning techniques can be used in this context that can help to improve the accuracy of the

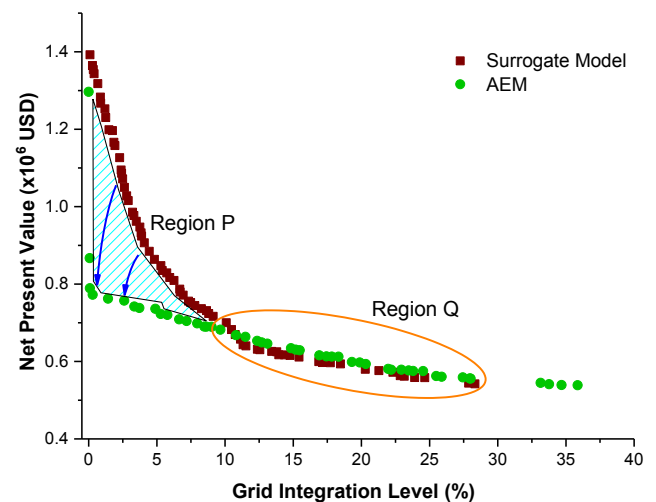


Fig. 7. Presents the two Pareto fronts obtained using Surrogate and AEM. Region Q represents the region where both Pareto fronts overlap with each other. Region P represents the sector which a significant deviation among the two Pareto fronts can be observed. Semi supervised techniques such as Active learning can be used to further improve the accuracy of the surrogate model by taking samples from Region P and further training the AEM.

Table 3

Comparison of Pareto solutions of the AEM and surrogate model obtained for grid interactions higher than 10%.

		NPV ($\times 10^6$ USD)	GI (%)	SPV capacity	Wind capacity	wind SPV	Battery banks	ICG capacity	ICG:non-dispatchable
A	A-AEM	0.6461	13.10	67.6	75	1.11	9	10	7.01
	A-S	0.6261	13.43	59.8	85	1.42	11	10	6.91
B	B-AEM	0.6132	17.44	67.6	70	1.04	8	10	7.27
	B-S	0.5966	17.73	63.7	90	1.41	10	7.5	4.88
C	C-AEM	0.5935	20.32	62.4	75	1.20	8	10	7.28
	C-S	0.5798	20.29	55.9	90	1.61	9	7.5	5.14
D	D-AEM	0.5605	25.89	58.5	65	1.11	8	10	8.10
	D-S	0.5582	24.64	58.5	85	1.45	10	10	6.97

surrogate model (Fig. 7).

6.2. Assessment of computational time for AEM, surrogate and hybrid models

The analysis conducted in Section 6.2 demonstrates that the surrogate model can approximate objective function values close to the Pareto solutions obtained using AEM in certain parts of the Pareto front. Furthermore, the surrogate model can provide a reasonable approximation for the decision space variables for a certain part of the Pareto front. However, it is clear that the surrogate model cannot be used alone to derive the Pareto solutions although it is efficient when computing the objective functions. The HOA introduced in this study becomes attractive in this context since it combines both surrogate and AEM. However, it is important to evaluate the boundary conditions valid for approximating the Pareto solutions using the surrogate model followed by AEM. When analyzing these Pareto fronts, the ratio of generation (REG) between AEM and surrogate model plays a vital role (Fig. 8). A higher REG (greater than 1) implies that the optimization algorithm uses the AEM for a larger number of generations than the surrogate model. Increasing the REG will extend the computational time since AEM takes longer for the computation. In contrast, AEM provides an accurate mapping of the decision space variables into the objective space. Hence, determining the optimum REG is important considering both convergence and computation time. Towards achieving this goal, a multi objective optimization is conducted considering different REG values. Among these scenarios, HOA 1 presents the case where the surrogate model is used for a larger number of generations when compared to HOA 2 and 3. Similarly, HOA 3 presents the case where AEM is used for a larger number of generations when compared to HOA 2 and 1. No significant difference can be seen when moving from one HOA to another in Fig. 8. More importantly, the objective function values of HOA 1 (which uses the surrogate model for a larger number of generations compared to HOA2 and 3) tend to follow the Pareto front obtained using AEM even when the grid integration levels are below 10%. Hence, it can be stated that the surrogate model

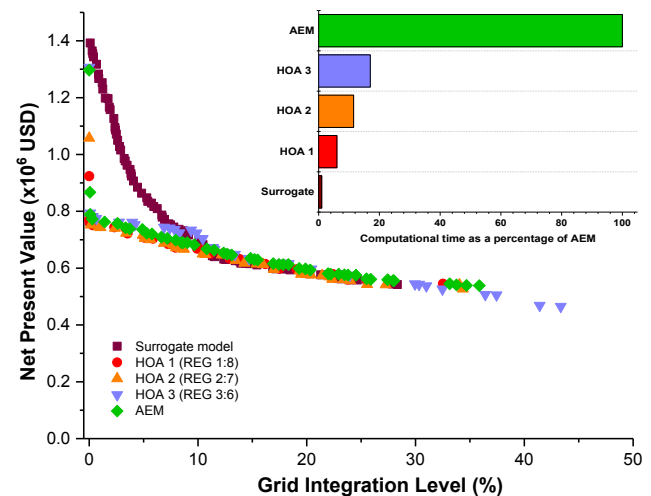


Fig. 8. Presents Pareto fronts obtained using different ratios of surrogate model to AEM. The three optimization algorithms with different combination ratios can reach the Pareto solutions in a fraction of the time taken by the AEM.

can be effectively merged with AEM using the HOA. Furthermore, the computational time decreases by 83% for the optimization process when using HOA 1, which is a significant achievement. However, it is interesting to assess the adaptability of the model for other cases where renewable energy potential and demand are different.

6.3. Comparison of the Pareto fronts for transfer-supervised learning

Sections 6.2 and 6.3 show the potential surrogate model to approximate the mapping of decision space variables into the objective space which can be used to minimize the computational time required for the energy system optimization. The main advantage of developing a surrogate model is its diverse applicability through modal adaptation. As discussed before, transfer learning facilitates adapting the surrogate

Table 4

Comparison of Pareto solutions of the AEM and surrogate model obtained for grid interactions less than 10%.

		NPV ($\times 10^6$ USD)	GI (%)	SPV capacity	Wind capacity	Wind : SPV	Battery banks	ICG capacity	ICG :non-dispatchable
E	E-AEM	0.6680	10.81	67.6	80	1.18	9	10	6.78
	E-S	0.7000	10.09	70.2	65	0.93	14	10	7.40
F	F-AEM	0.7204	5.77	68.9	85	1.23	10	10	6.50
	F-S	0.8167	5.95	71.5	45	0.63	20	10	8.58
G	G-AEM	0.7420	3.35	75.4	80	1.06	11	10	6.44
	G-S	0.9772	3.32	58.5	40	0.68	20	12.5	12.69
H	H-AEM	0.7573	2.60	75.4	85	1.13	11	10	6.23
	H-S	1.1578	1.93	55.9	25	0.45	20	15	18.54
I	I-AEM	0.7623	1.42	68.9	85	1.23	10	10	6.50
	I-S	1.1986	1.49	81.9	0	0.00	20	25	30.53

model developed using supervised learning. In order to assess the model adaptation capability, six different scenarios are taken where the renewable energy potential for solar PV (SPV 1 and SPV 2), wind (Wind 1 and Wind 2) and demand (Demand 1 and Demand 2) profiles are different from one another. AEM is used to generate the Pareto solutions for different scenarios as the first step of the analysis (Fig. 9).

The changes observed in the Pareto fronts in Fig. 9 are the consequences of the changes introduced in demand, solar and wind energy potential. It is interesting to assess the capability of transfer learning to adapt the surrogate model trained for the initial conditions. When comparing the Pareto fronts obtained using AEM, it is prudent that SPV 1 and 2 closely follow the Initial Scenario. However, notable changes in the Pareto fronts are observed for Scenarios Demand 1, 2 and Wind 1, 2. For example, the shape of the Pareto front notably changes when moving from initial scenario to Wind 2 as a consequence of the changes introduced in the scenario. It is interesting to evaluate the adaptability of the surrogate model through transfer learning (SMTL). Towards this end, Pareto fronts obtained entirely using SMTL are compared with AEM for different scenarios (Fig. 10). When comparing the Pareto fronts obtained using AEM and SMTL, it is clear that SMTL provides a good approximation when considering the objective function values for all scenarios. For example, Pareto fronts obtained using SMTL and AEM for scenarios SPV 1 and 2 closely follow each other when minimizing the grid interactions until a grid interaction level of 10% is reached (similar to the Pareto front obtained using AEM and the surrogate model for Initial Scenario) (Fig. 10). Demand 1 Scenario introduces a significant change to the Pareto front obtained using AEM (Fig. 9). Nonetheless, SMTL provide a better approximation to the AEM for the scenario Demand 1. When considering the scenario Wind 2, Pareto front solutions obtained using AEM show a complex variation in the objective space when compared to the Initial Scenario. However, it is observed that the SMTL has the potential to closely follow the AEM when decreasing grid interactions to 6%. In conclusion, it can be stated that the objective function values obtained using SMTL alone provide a better approximation, similar to the surrogate model trained using supervised learning. Hence, SMTL becomes a better substitute for the surrogate model and can be effectively used whenever boundary conditions are changed.

Similar to the surrogate model, SMTL is combined with AEM by using HOA (SMTL replaces the surrogate model) and found that REG of 1:8 is sufficient for the Pareto front to be similar to the initial scenario where the surrogate model is trained using supervised learning. This makes it interesting to analyze the entire computational time required to train the computational model and conduct the optimization. The computational time includes the time required to generate the data set for transfer learning, computational time for transfer learning and optimization (Fig. 11). When compared with the optimization algorithm based on AEM, a significant drop in computational time can be observed when using SMTL along with the HOA (REG of 1:8). The computational time required all together for generating the data points using the AEM for transfer learning, transfer learning for the model and subsequently optimizing the energy system takes 15.3% of the computational time required for the AEM. Such a significant drop in the computational time required for energy system optimization will enable optimizing energy systems at regional and national scale consisting of a large number of distributed energy systems.

7. Conclusions

Distributed energy systems play a vital role in the renewable energy integration process. Optimum design of distributed energy systems is a computationally intensive process which may take several days in certain instances. Furthermore, the optimization process is not flexible enough to adapt to changes in renewable energy potential, demand and other input data. As a result, an optimization algorithm needs to be implemented from the beginning, when there are changes in the

aforementioned factors. Therefore, it is challenging to evaluate the potential for distributed energy systems in the regional and national scale using bottom up models.

Supervised and transfer learning techniques have been used in this study to improve the computational speed and model adaptation. Supervised learning technique is used in this study to develop a surrogate model to assist energy system optimization. Besides being limited to system configuration, as it is practiced in present, both system configuration and dispatch strategy are considered when developing the surrogate model. Subsequently, both surrogate model and actual

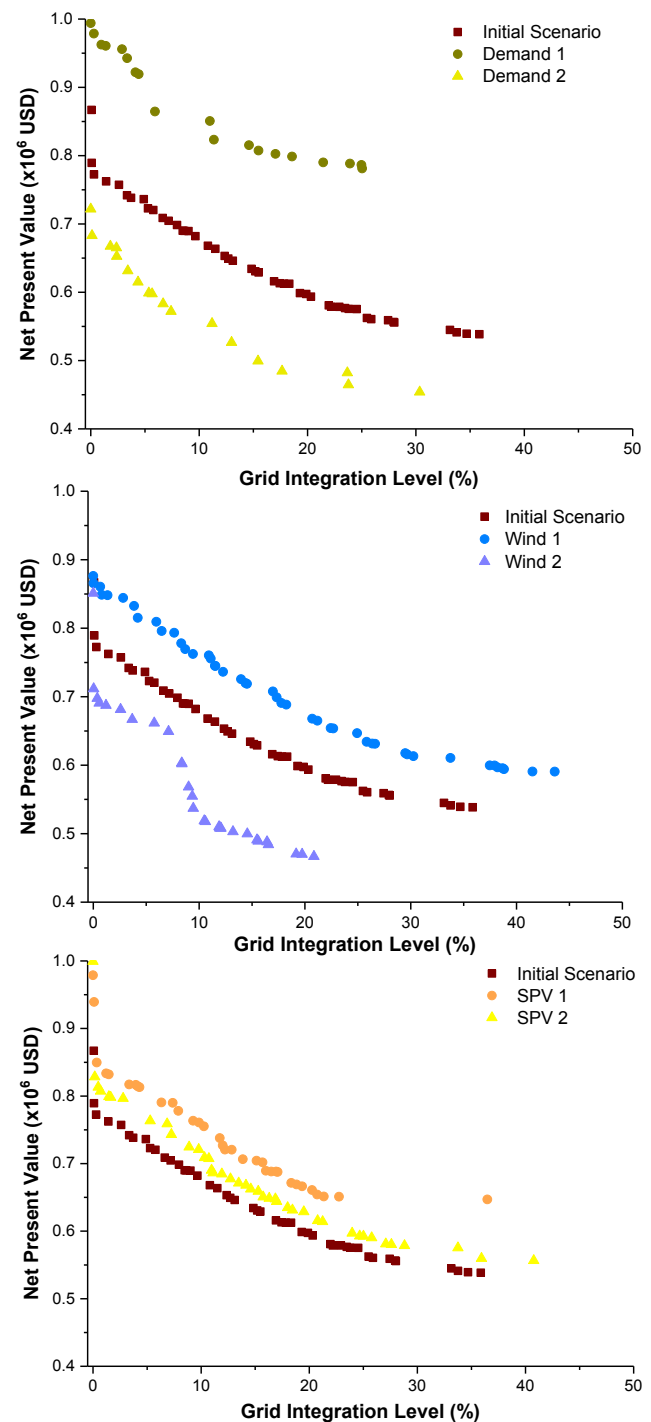


Fig. 9. Pareto fronts obtained using the AEM for six different scenarios are presented from top to bottom. A notable shift in Pareto fronts is observed for wind and demand scenarios.

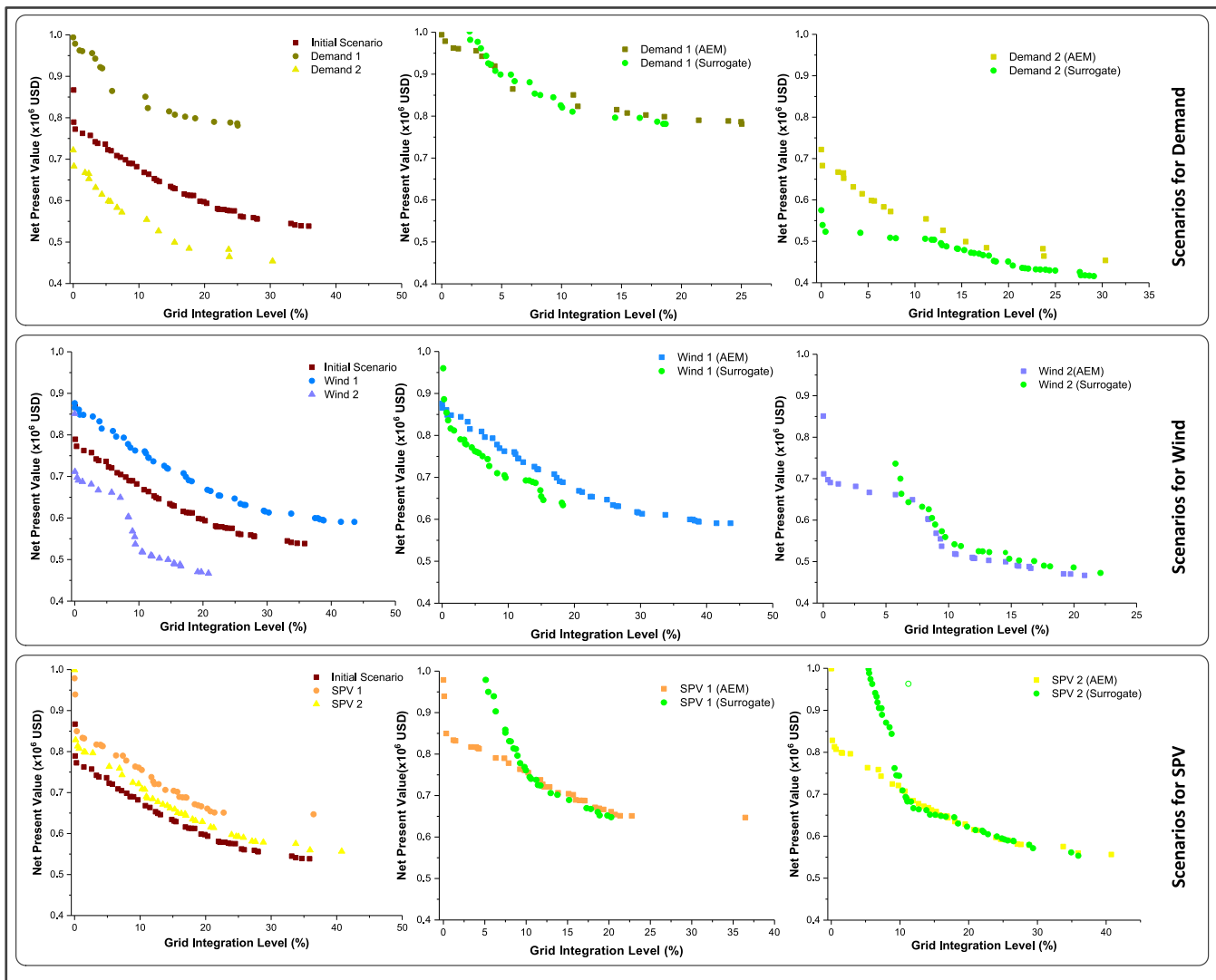


Fig. 10. Pareto fronts obtained using the surrogate model that is trained using transfer learning (SMTL) technique for a data set consisting of 10,000 data points. The Pareto fronts clearly demonstrate that transfer learning technique has a good potential to adapt the model.

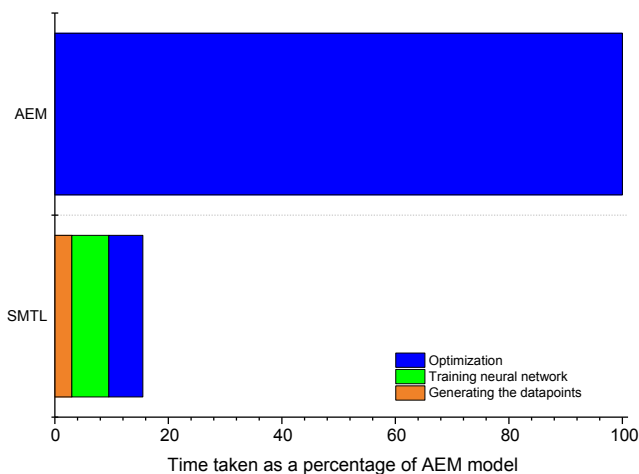


Fig. 11. Comparison of the computational time required for the classical approach based on AEM and the novel approach based on SMTL. A notable reduction in computation time attained by using SMTL which enables this method to be used to optimize large pool of distributed energy systems at regional and national scale.

engineering model are linked with the novel optimization algorithm introduced in this work. The developed surrogate model shows the potential to approximate the objective function values of the Pareto solutions with a higher efficiency. However, there is a significant deviation in system design for the Pareto solutions obtained using the surrogate model. When analyzing the Pareto solutions further, it is revealed that the surrogate model possesses the potential to predict the general trend of renewable energy components, battery bank and Internal Combustion Generator. However, it fails to predict the changes in system operation variables. This specific weakness can be improved using active learning which will be an extension to the present study. The surrogate model can be effectively combined with the Actual Engineering Model using the novel optimization algorithm introduced in this study (Hybrid Optimization Algorithm). The novel computational algorithm can reduce the computational time of the optimization process significantly (by up to 84%).

The surrogate model initially trained for one specific scenario is later adapted using transfer learning for six different scenarios where solar, wind and demand profiles are notably different. A data set (10,000 data points) is generated using the Actual Engineering Model to train the surrogate model. Surrogate Model based on Transfer Learning is developed with the assistance of the initial surrogate model through transfer learning. Surrogate Model based on Transfer Learning has the

potential to approximate the Pareto solutions with the support of an optimization algorithm with a similar accuracy achieved by the surrogate model trained using supervised learning. Hybrid Optimization Algorithm that combines the surrogate model and Surrogate Model based on Transfer Learning can reach the Pareto solutions within a fraction of time taken by Actual Engineering Model, reducing the computational time by 84%. Such a significant reduction in computational time facilitates use of the proposed method for evaluating the potential of a large number of distributed energy systems at the regional and national scales.

This study reveals that it is possible to minimize the computational time required for energy system optimization though adopting machine learning technique. However, it is important to derive the energy demand and renewable generation potentials in order to conduct energy system optimization at the regional and national scales. Most of the models for estimating the energy demand and renewable generation potential consider daily or monthly temporal resolutions. Hence, it is important to move into hourly time resolution in order to match with the energy system optimization model developed in this study. Machine learning techniques can be effectively used in this context to provide energy demand and generation data for the regional and national scales.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2019.03.202>.

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